16. On Series Expansion in Form Theory

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By the "series expansion of form theory" one understands, as is well known, on the one hand the generalization of the Clebsch–Gordan series expansion – that is, the expansion of a form containing two binary rows according to powers of the determinant of these rows, where the coefficients are polars of forms in only one row of variables, or, what is identical with this, vanish under application of the Ω-process – to forms in arbitrarily many series of n variables each; and, on the other hand, as a generalization of the normal forms occurring for the "complex coordinates" t_{ik}, an expansion according to normal forms for forms which contain simultaneously the quantities of different levels $t_i, t_{ik}, t_{ikl}, \ldots$. This second kind of expansion can, however, be obtained from the first by invariant differentiation processes, as was carried out above all by Mertens and Deruyts.[1]

The various expansions of the first kind all arise – as is briefly sketched, for instance, in Math. Ann. 77, loc. cit. III – as consequences of the expansion of a form in n rows of n variables each according to powers of the determinant of these rows, where the coefficients are polars of forms in only $(n-1)$ rows, which themselves are derived from the initial form by polar formation and the Ω-process.[2] The original Clebsch–Gordan expansion $(n = 2)$ had gone one step further: the special polar processes which occur there are given explicitly, even including the numerical coefficients.

What I now wish to add to the indicated note – where the series expansion is conceived more conceptually, as a problem of linear pencils of forms – is an explicit representation up to numerical coefficients, obtained by specializing E. Fischer's investigations concerning general module systems.[3]

This classification is led to by interpreting the normal form in the t_{ik}. Namely, if one replaces the "complex coordinates" t_{ik} by the "line coordinates"

$$p_{ik} = (x_i y_k - x_k y_i),$$

then, because of the identical relations existing among the p_{ik}, the representation of a form $F(p_{ik})$ is no longer unique. Uniqueness can be obtained by requiring that among the coefficients of F there hold the same linear relations as among the power products of the p_{ik} occurring in F.[4] Thus for this normal form Φ one has:

$$\begin{aligned} F(p_{ik}) &= \Phi(p_{ik}) \quad [\text{identically in } x, y], \quad \text{or} \\ F(t_{ik}) &\equiv \Phi(t_{ik}) \quad \bmod M, \end{aligned} \tag{1}$$

where M denotes the module of all identical relations among the p_{ik}, that is, consists of the totality of all forms $G(t_{ik})$ for which $G(p_{ik})$ vanishes identically in x, y. $\Phi(t_{ik})$ is the "normal form" in complex coordinates, while the basis functions of M become quadratic forms in the t_{ik}. By iterating the procedure with respect to the factors of these basis functions, the complete series expansion arises. Corresponding considerations hold when $t_i, t_{ik}, t_{ikl}, \ldots$ occur simultaneously.

By an analogous line of thought one can reduce the expansion according to polars. If, for example, in a form $F(x, y, z)$ one replaces the three ternary rows $x = (x_1, x_2, x_3), y, z$ by rows ξ, η, ζ which are linearly composed of only two of them (for instance $\xi = x, \eta = y; \zeta = \lambda_1 x + \lambda_2 y$), then the representation of $F(\xi, \eta, \zeta)$ is again not unique. One can again prescribe for the coefficients the same linear relations as for the power products of ξ, η, ζ occurring in F. Since here the module M of all identical relations has as basis function the determinant $\Delta = (xyz)$ of the three rows (loc. cit. III, auxiliary lemma a)), [1] is replaced by:

$$\begin{aligned} F(\xi, \eta, \zeta) &= \Phi(\xi, \eta, \zeta) \quad [\text{identically in } x, y], \quad \text{or} \\ F(x, y, z) &\equiv \Phi(x, y, z) \quad \bmod \Delta. \end{aligned} \tag{2}$$

It follows that Φ arises by polar formation (compare formula (2) loc. cit.); the complete series expansion is again obtained by iteration.

[1] F. Mertens, Über invariante Gebilde quaternärer Formen (Sitzber. d. Ak. d. Wiss. Wien 98, Abt. IIa (1889)) for the quaternary case. The general normal forms are found in J. Deruyts: Sur la réduction des fonctions invariantes dans le système des variables géométriques, Bulletins de l'Ac. R. de Belgique 24 (1892). Without knowing this work, and in close dependence on Mertens, I gave the general expansion according to normal forms in § 7 of the paper: Zur Invariantentheorie der Formen von n Variabeln, J. f. M. 139 (1910).

[2] Compare the bibliographical references loc. cit. Since then there has appeared, also based on the formal method: Schouten, Over reeksontwikkelingen van algebraische vormen met verschillende rijen van variabelen van verschillenden graad, Verslag Ak. Amsterdam 27 (1919), p. 1481.

[3] E. Fischer, Über die Differentiationsprozesse der Algebra, J. f. M. 148 (1917).

[4] Compare Clebsch, Über eine Fundamentalaufgabe der Invariantentheorie, § 4, Abh. d. K. G. d. Wissensch. zu Göttingen 17 (1872).

The formation of the normal forms Φ is therefore subordinated in both cases to the following problem: if in $F(z)$ the indeterminates z_i are replaced by polynomials $f_i(\xi)$ in parameters ξ, so that the representation $F(f_i(\xi))$ is therefore no longer unique, then the uniqueness is to be fixed by requiring that among the coefficients there hold the same linear relations as among the power products of the $f_i(\xi)$ occurring in F. Thus one has

$$\begin{aligned} F(f_i(\xi)) &= \Phi(f_i(\xi)) \quad \text{[identically in } \xi], \quad \text{or} \\ F(z) &\equiv \Phi(z) \qquad\qquad \operatorname{mod} M, \end{aligned} \tag{3}$$

where M denotes the module of all identical relations among the $f_i(\xi)$.

For this normal form $\Phi(z)$, E. Fischer, in § 11, I of the cited paper, gave an explicit expansion up to numerical coefficients, namely by means of linear operations. At the same time, however, in Introduction III and in § 6 II, he also gave, for the difference $F - \Phi$ belonging to M – assuming knowledge of a module basis – an explicit expression in the same sense; and, by combining these developments in § 11 II, he thereby replaced formula [3] by an explicit representation. The question of the numerical coefficients, still open here, is according to § 12 of the same paper identical with the determination of the eigenvalues and eigenfunctions of the linear operations (formula 82).

I now specialize Fischer's formulas, in 1., to the case of the series expansion according to polars, and obtain from them, by iteration, the series expansion. The operation for determining Φ thereby passes into definite polar processes, which is considerably more precise than the previously known formulation given above. In the operation for determining $F - \Phi$, multiplication by the determinant Δ and the Ω-process occur in combination. To get to the usual, non-explicit form, one must also note that $\Omega\Delta$ can be expressed by polar processes.

At this point a correction should also be added. In the work cited above (p. 101, line 6 from the top) I inadvertently assumed as generally valid the identical transformation

$$\Omega(\Delta\psi) = c_1\psi + c_2\Delta \cdot \Omega(\psi),$$

which holds only in the binary domain, and thereby made the passage from the fundamental formula (2), which gives the expression for the normal form Φ in [2], to the series expansion (3). The considerations there therefore yield only formula (2), and as a special case of it the reduction theorem (1); but, since an explicit expression for the difference $F - \Phi$ is missing, they do not suffice to justify the series expansion.

In 2. I similarly obtain, by specialization, the normal form $\Phi(t_{ik})$ of [1]; more generally, the normal form $\Phi(t_i, t_{ik}, t_{ikl}, \ldots)$, which leads to known explicit formulas. But since setting up the complete series expansion requires knowing a module basis of M, the invariant-theoretic route (compare the cited literature) is preferable here; it does not require this knowledge, but on the contrary supplies a module basis by means of the series expansion.

1. According to Fischer, § 11 (formula (19) and the following lines), the normal form $\Phi(z)$ of [3] – there denoted $N(\zeta)$ – is given by:

$$\Phi(z) = (a_1 S + a_2 S^2 + \cdots + a_r S^r)F(z), \tag{4}$$

where the constants $a_1 \ldots a_r$ belong to the coefficient domain of the $f(\xi)$ and depend only on the degree of $F(z)$; the operator S is defined by

$$\begin{aligned} S = AB; \qquad BF(z) &= F(f(\xi)); \\ AF(f(\xi)) &= \left\{\sum z_i f_i\left(\frac{\partial}{\partial \xi}\right)\right\}^p F(f(\xi))^5. \end{aligned} \tag{5}$$

If in [2] one regards F as a form of p-th dimension in z alone, then $f_i(\xi)$ specializes to $\lambda_1 x_i + \lambda_2 y_i$; consequently one obtains:

$$BF = F(x, y, \lambda_1 x + \lambda_2 y),$$

$$SF = \left\{\sum z_i \left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_i} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_i}\right)\right\}^p F(x, y, \lambda_1 x + \lambda_2 y),$$

or, written in polar processes (compare loc. cit. II, 2 and 3; also for the abbreviated notation):

$$SF(x, y, z) = \frac{1}{p!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^p \left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^p F(x, y, z). \tag{6}$$

[5] Compare formula (75). In the case of [3], the sum α runs over all power products of p-th dimension, and (75) therefore specializes to [5].

Equations [4] *and* [6] *give the desired explicit expression for* $\Phi(x,y,z)$; at the same time $\Phi(x,y,z)$ falls under the general form mentioned at the start ($\sum PH$ in (2), loc. cit.) – polars of expressions which depend on only 2 rows and are obtained from F by polar processes. The still undetermined coefficients a_i are rational numbers, since here the $f_i(\xi)$ have integral-rational numerical coefficients. If [2] is written in the form

$$F = \Phi + \Delta F_1, \tag{2a}$$

then specialization of the formula given by Fischer in Introduction III and in § 6 II (p. 41) gives

$$F_1 = (c_1\Omega + c_2\Omega\Delta\Omega + \cdots + c_i\Omega\Delta\Omega\cdots\Delta\Omega)F^6, \qquad \left[\Omega = \left(\frac{\partial}{\partial x}\frac{\partial}{\partial y}\frac{\partial}{\partial z}\right)\right], \tag{7}$$

where the c_i are again rational numbers depending only on the degree of F.

The iteration leads to:

$$F_1 = \Phi_1 + \Delta F_2; \quad F_2 = \Phi_2 + \Delta F_3; \quad \ldots; \quad F_i = \Phi_i + \Delta F_{i+1}; \quad \ldots,$$

where in each case the Φ_i are the normal forms corresponding to the F_i. Since the F_i have degree $(p-i)$ in z, [4] and [6] give:

$$\Phi_i = (a_{i1}S_i + a_{i2}S_i^2 + \cdots)F_i,$$
$$S_iF_i = \frac{1}{(p-i)!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^{p-i}\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^{p-i} F_i. \tag{8}$$

Formula [7] corresponds to

$$F_i = (\alpha_{i1}\Omega + \alpha_{i2}\Omega\Delta\Omega + \cdots)F_{i-1}, \quad \text{and by iteration}$$

$$F_i = (\beta_1\Omega^i + \beta_2\Omega^i\Delta\Omega + \cdots + \beta_\rho\Omega^{k_1}\Delta\Omega^{k_2}\cdots\Delta\Omega^{k_\lambda} + \cdots)F, \quad (k_1 + k_2 + \cdots + k_\lambda - \lambda + 1 = i). \tag{9}$$

By means of [8] *and* [9], *the expansion*

$$F = \Phi + \Delta\Phi_1 + \Delta^2\Phi_2 + \cdots + \Delta^i\Phi_i + \cdots + \Delta^p\Phi_p$$

has reached an explicit representation in the stated sense.

If one applies to [7] the identical transformation $\Omega\Delta F = PF$, where P denotes polar processes,[7] then, since these processes commute with the Ω-process, one obtains the known formula

$$F_1 = P\Omega F; \quad \ldots; \quad F_i = P\Omega^i F;$$

and from this, in conjunction with [8] or with the considerations loc. cit., follows the known formulation of the series expansion, as for instance in (3) loc. cit. If one is dealing with n rows of n variables, $x^{(1)}, x^{(2)}, \ldots, x^{(n-1)}, z$, then correspondingly

$$f_i(\xi) = \lambda_1 x_i^{(1)} + \lambda_2 x_i^{(2)} + \cdots + \lambda_{n-1}x_i^{(n-1)};$$

and likewise Δ and Ω are to be defined for n rows.

2. For the normal form of $F(t_i, t_{ik}, t_{ikl}, \ldots)$, the functions $f_i(\xi), f_{ik}(\xi), \ldots$ are given by the one-row, two-row, up to $(n-1)$-row determinants

$$\xi_i^{(1)}, \quad (\xi^{(1)}\xi^{(2)})_{ik}, \quad \ldots, \quad (\xi^{(1)}\xi^{(2)}\cdots\xi^{(n-1)})_{i_1 i_2\cdots i_{n-1}}.$$

Thus, as a specialization of [5], one obtains:

$$S = AB; \qquad BF = F(\xi_i^{(1)}, (\xi^{(1)}\xi^{(2)})_{ik}, \ldots),$$
$$ABF = \left(\sum t_i\frac{\partial}{\partial\xi_i^{(1)}}\right)^{\lambda_1}\left(\sum t_{ik}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\right)_{ik}\right)^{\lambda_2}\cdots F(\xi_i^{(1)}, (\xi^{(1)}\xi^{(2)})_{ik}, \ldots),$$

where $\lambda_1, \lambda_2, \ldots$ denote the degrees in $t_i, t_{ik}, \ldots$.

[6] Kerim uses this specialized formula in his Erlangen dissertation, soon to appear, in the application to "inertia forms".

[7] Compare, for example, F. Mertens, Über ganze Funktionen von m Systemen von je n Unbestimmten, Monatsh. f. Math. u. Phys. 4, formula (5).

From invariant-theoretic considerations (namely, all invariants of F which are linear in the coefficients can be linearly composed from Φ and from forms out of M, thus in particular also the SF not belonging to M)[8] the congruence follows here: $\Phi \equiv a_1 SF \mod M$; and because of the normal-form property one obtains from this, instead of [4], the simple relation

$$\Phi = a_1 SF = a_1 S\Phi. \tag{10}$$

The second equality holds because applying S makes every form from M vanish. Formula [10] shows that the normal form Φ is itself at the same time an eigenfunction; and indeed, by the invariant-theoretic consideration above, there are no further eigenfunctions which are at the same time invariants of F (compare, concerning [10], Deruyts, formulas (10) and (10')).

Corrections.

1. In the paper "Die allgemeinsten Bereiche aus ganzen transzendenten Zahlen" (Math. Ann. 77), on p. 121, line 8 from the top, line 9 from the top, and line 11 from the bottom, the field (H, Y) is to be replaced by the "field $(H, Y, Y', \ldots, Y^{(\lambda_0-1)})$, where $Y', \ldots, Y^{(\lambda_0-1)}$ denote the conjugates of Y with respect to H." Correspondingly, on p. 120, line 5 from the top, and p. 121, line 9 from the top, (H, y) is to be replaced by $(H, y, y', \ldots, y^{(\lambda_0-1)})$.

2. In the paper "Gleichungen mit vorgeschriebener Gruppe" (Math. Ann. 78), in the formulation of Hilbert's irreducibility theorem – p. 222, line 3 from the bottom – instead of "number field Ω" it must more precisely say "finite algebraic number field Ω", as Mr. Ostrowski has pointed out to me. In fact, with respect to the field of all algebraic numbers, for instance, the group of every equation with algebraic numerical coefficients is the identity group. – Correspondingly, in the introduction, "arbitrary domain of rationality" is to be understood as a finite algebraic number field to which finitely many parameters can still be adjoined, or as one of the intermediate fields which arise in this way.

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[8] This follows, for instance, from § 6 and § 7, 2. of my cited paper "Zur Invariantentheorie der Formen von n Variabeln".